Auditing the Faithfulness and Cross-City Consistency of Attention in Multimodal Spatiotemporal Air Quality Index Prediction

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***Abstract*—Deep learning models for Air Quality Index (AQI) prediction increasingly embed attention mechanisms, and the resulting attention weights are routinely presented as explanations of what the model has learned about pollution. However, an attention weight is not equivalent to a predictive or causal contribution, and this assumption is seldom tested in the air-quality domain, which is dominated by regression targets, physically correlated inputs, and spatiotemporal structure that differ sharply from the natural-language classification settings where the "attention as explanation" debate originated. This work develops a region-agnostic multimodal spatiotemporal AQI prediction model, trained and evaluated across multiple cities that span distinct climatic and emission regimes, fusing satellite retrievals (NO2, HCHO, AOD), meteorological reanalysis, geographic location, and AQI history, with attention operating over environmental features and temporal observations. The model is benchmarked against classical, sequence, and transformer baselines and against attention-ablated variants, and is evaluated both within cities and under a leave-one-city-out protocol that measures cross-region transfer. The central research question is whether the attention mechanism provides a faithful representation of the factors that actually drive the model's predictions, and whether that faithfulness is consistent across cities. Attention weights (raw and flow-corrected) are compared against SHAP, LIME, grouped permutation importance, and concept activation testing, using established faithfulness metrics—comprehensiveness, sufficiency, deletion curves, and remove-and-retrain—and against known atmospheric relationships. All analyses are conditioned on pollution regime, season, city, and climatic zone, with emphasis on extreme-pollution episodes. Expected outcomes are a characterization of when and where attention is faithful, a physics-adjudicated protocol for resolving disagreements among explanation methods, a measurement of how explanation faithfulness transfers across regions, and a concept-level test of model sensitivity to atmospheric stagnation. The contribution is an explainability audit rather than a new predictor.**

***Index Terms*—Air quality index, explainable AI, attention mechanism, faithfulness, SHAP, LIME, spatiotemporal deep learning, remote sensing, cross-region generalization, concept activation vectors, atmospheric stagnation.**

I. Introduction and Motivation

Air quality varies steeply in space and time and is governed by the interaction of emissions, atmospheric composition, and meteorology. These drivers differ markedly between cities: an inland basin subject to winter stagnation, a coastal city with strong sea-breeze ventilation, and a semi-arid city dominated by dust do not share the same pollution physics. Machine-learning models that combine satellite-derived pollution proxies with meteorological reanalysis and ground-station measurements can predict AQI at fine resolution across many such settings, and attention-based architectures are now common because they are accurate and are assumed to be interpretable: the attention distribution is read as a ranking of the environmental variables and past time steps the model considered important.

This assumption is questionable. A substantial body of work outside the environmental domain has shown that attention weights can be altered without changing predictions, correlate only weakly with gradient- and perturbation-based importance, and are not, in general, faithful explanations. That evidence comes almost entirely from text classification. Whether it transfers to a continuous AQI regression target, driven by strongly correlated physical inputs and lagged observations, with attention spanning multiple data modalities and time, and whether any such behaviour is stable across cities with different pollution regimes, are open questions. Air-quality studies that pair attention with post-hoc methods such as SHAP typically report the two side by side for a single city and treat agreement

as validation, without measuring faithfulness, testing consistency across regions, or explaining disagreement.

This project therefore treats "attention weight approximates predictive importance" as a hypothesis to be tested, not a property to be assumed, in a multi-city multimodal spatiotemporal AQI model, and asks whether the answer is the same everywhere.

II. Problem Statement and Research Questions

Given a model that predicts daily station AQI for a set of cities from satellite, meteorological, spatial, and historical inputs and exposes attention over features and time, the study asks:

RQ1. Do high-attention features also receive high importance from SHAP, LIME, and grouped permutation/perturbation analysis?

RQ2. Does perturbing high-attention features change the prediction more than perturbing low-attention features, as measured by comprehensiveness, sufficiency, deletion curves, and remove-and-retrain?

RQ3. Are attention-derived rankings stable across AQI levels, across seasons, and across individual monitoring stations within a city?

RQ4. Does attention remain faithful during extreme-pollution episodes, or does faithfulness degrade in the high-AQI tail?

RQ5. Where, and by how much, do attention and independent explanation methods disagree, quantified with feature-, rank-, and sign-agreement metrics?

RQ6. When methods disagree, which is more consistent with established atmospheric relationships (pollutant dispersion with wind speed, accumulation under stagnation, boundary-layer venting, hygroscopic growth with humidity)?

RQ7. Are attention rankings and their faithfulness consistent *across cities and climatic regimes*, and do the explanations produced for a held-out city—one absent from training—remain faithful and physically plausible?

RQ8. Is the model's internal representation sensitive to the physically defined concept of atmospheric stagnation during high-AQI predictions, and does that sensitivity agree with feature-level attention and SHAP across regimes?

A secondary question (**RQ0**) is whether each data source and architectural component—satellite, meteorology, spatial encoding, temporal history, and attention—measurably improves prediction, both within cities and across cities.

III. Proposed Methodology

A. Study regions and data

The study uses a set of cities chosen to span distinct climatic and emission regimes (for example, plains cities prone to winter stagnation, coastal cities with strong ventilation, and semi-arid dust-influenced cities). Inputs are satellite retrievals (NO_2, HCHO, AOD), meteorological reanalysis (2 m temperature, 2 m dewpoint, surface pressure, 10 m wind components, total precipitation, boundary-layer height, with derived relative humidity and wind speed/direction), and geographic coordinates. Ground truth is national air-quality-agency station AQI. Satellite, meteorological, and station records are aligned to a common location and daily time step, and agency sub-indices are computed with the official formula. Each sample is tagged with its city, climatic zone, AQI band, and season.

B. Prediction models

A spatiotemporal model consumes a window of past observations and applies attention across environmental features and across time steps before a regression head, with the city/location supplied as context so that a single model serves all regions. Baselines span linear regression, random forest, support vector regression, gradient boosting, a multilayer perceptron, LSTM/GRU recurrent networks, and a transformer / temporal-fusion-transformer. An ablation ladder—satellite only, then adding weather, spatial encoding, temporal history, and finally attention—isolates each component; an attention-free variant of the proposed model is a direct control.

C. Evaluation protocols

Two regimes are used. *Within-city*: temporally ordered train/validation/test splits per city, with no look-ahead leakage. *Cross-city*: leave-one-city-out, training on all but one city and evaluating prediction and explanations on the held-out city, to measure geographic transfer of both accuracy and faithfulness.

D. Explanation methods under comparison

For each test prediction the study extracts attention weights (raw and aggregated with attention rollout/flow), SHAP values, LIME attributions, and grouped permutation/perturbation importance that respects the correlation structure of the physical inputs. For higher-level analysis, concept activation vectors are trained for "atmospheric stagnation," defined from air-stagnation-index components (low 10 m wind, low boundary-layer height, low precipitation).

E. Faithfulness, agreement, and physical adjudication

Faithfulness is measured with comprehensiveness and sufficiency (area over the perturbation curve), deletion/insertion curves, and remove-and-retrain, which controls for perturbation-induced distribution shift. Cross-method agreement is quantified with feature-, rank-, sign-, and signed-rank-agreement metrics. All measures are reported globally and stratified by city, climatic zone, AQI band, and season. Disagreements among methods are scored against a checklist of established atmospheric relationships, so that physical plausibility is measured rather than asserted.

IV. Expected Contributions

1) A systematic faithfulness audit of attention in a multi-city multimodal spatiotemporal AQI regression model, using standard faithfulness metrics rather than informal ranking comparison.

2) The first measurement, to our knowledge, of how explanation faithfulness and attention rankings transfer across cities and climatic regimes, including to cities unseen during training.

3) A protocol that adjudicates disagreement among explanation methods against atmospheric physics, converting the known

XAI disagreement problem into an evaluable question in this domain.

4) A characterization of the conditions—regime, season, location, and extreme episodes—under which attention is or is not a faithful explanation.

5) A concept-level test of model sensitivity to atmospheric stagnation, extending explanation from individual features to a physically meaningful environmental concept, and checking whether that sensitivity is consistent across regions.

6) An ablation study that reports explanation faithfulness alongside predictive accuracy, testing whether components that lower error also yield more physically coherent explanations.

V. Evaluation Plan

Predictive performance is evaluated with MAE, MSE, RMSE, and R^2, reported overall, per city, per AQI category, and for identified high-pollution episodes, under both the within-city and leave-one-city-out protocols. Explanation quality is evaluated independently of accuracy, using the faithfulness and agreement metrics above together with the physical-consistency score, and is compared between seen and held-out cities. The study explicitly allows for negative or mixed results: a finding that attention is largely unfaithful, or largely faithful, or faithful only in some regimes, is informative provided the evaluation is rigorous and the operating conditions are characterized. Metrics and decision thresholds are fixed in advance.

VI. Scope and Limitations

The study covers a bounded set of cities and a defined multi-year period at daily temporal resolution, using the satellite products and meteorological fields listed above and national-agency station AQI as ground truth. Assembling aligned multi-city data is a prerequisite and a substantial part of the work. Additional pollutants (CO, SO_2, O_3), land-use and elevation layers, sub-daily forecasting, generalization beyond the studied country, and formal causal inference beyond perturbation are outside the present scope and are identified as future work.

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